

ML Assignment-1

Pennam Nikhil - IMT2024070

GitHub Link: <https://github.com/Nikhil85587/ML-Assignment-1>

1.Introduction:

1.1 My understanding of the problem statement:

The objective is to select a polynomial regression model that achieves good validation performance while avoiding underfitting and overfitting. There were 6 features in the first question and 3 features in the second question. We had to consider polynomials until degree 10 for the first question, degree 20 for the second question and finally choose a polynomial that is the best fit for these data points while making sure that there is no overfitting.

1.2 My Approach:

So, the first thing I did was to separate X and y to prevent data leakage while training. I then tried polynomials from degree 1 to degree 10 for the first question and degree 1 to degree 20 for the second question. For each degree, I evaluated both L1 and L2 regularization with different α values. The polynomial features were standardized using StandardScaler before applying regularization.

I used K-fold validation for selecting the best model. For each combination of degree, regularization type, and alpha(0.0001, 0.001, 0.01, 0.1, 1, 10, 100), I trained the model on 80% of the data and used the remaining 20% as validation data to calculate R^2 and MSE. This was done on 5 different splits (I used K=5), and the average was taken for each combination. I then selected the combination with the lowest mean MSE. R^2 was also noted to assess model performance. After the final model was selected, I used the whole training data to train it again. The resulting regularized polynomial model was then used for the final prediction.

2.Methodology

2.1 Data Preparation

The first step was to separate the input features from the target variable. The input features were stored in X, while the target variable yy was kept separately. This ensures that the target variable is not inadvertently used as an input feature during model training. For Question 1, all six available features, x1, x2, x3, x4, x5, and x6, were used. For Question 2, all three available features, x1, x2, x3, were used.

The training and test datasets were kept separate throughout the modelling process. The test dataset was not used during polynomial degree selection or cross-validation. It was used only after the final model had been selected and retrained. This separation helps prevent data leakage, where information from the test dataset could unintentionally influence the model-selection process.

2.2 Polynomial Feature Generation

Polynomial regression was implemented by transforming the original input features into polynomial features using the PolynomialFeatures method from Scikit-learn. The transformed features are then provided to a linear regression model.

Different polynomial degrees were evaluated to determine an appropriate model complexity.

- Question 1: Polynomial degrees from 1 to 10
- Question 2: Polynomial degrees from 1 to 20

Evaluating multiple degrees allows the model's validation performance to be compared as its complexity increases.

2.3 Polynomial Regression Model

For each polynomial degree, the input data was first transformed into polynomial features. A standard Linear Regression model was then fitted on these transformed features. The model complexity was controlled by selecting an appropriate polynomial degree based on cross-validation performance.

2.4 Model Validation

To evaluate the different polynomial degrees, 5-fold cross-validation was used. In 5-fold cross-validation, the training dataset is divided into five approximately equal subsets. For each iteration, four subsets are used for training and the remaining subset is used for validation. Seed number 42 has been used to keep these splits consistent

Therefore, each iteration uses approximately:

- 80% of the training data for model fitting
- 20% of the training data for validation

This process is repeated five times so that each subset is used as the validation set once.

For every polynomial degree, two evaluation metrics were calculated on the validation data:

Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

A lower MSE indicates better validation performance.

R² Score

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

The R² score measures how much of the variation in the target variable is explained by the model. A higher R² generally indicates better predictive performance.

After completing all five folds, the mean of the MSE and R² values were calculated for each polynomial degree. The mean values provide an overall estimate of validation performance, while the standard deviation indicates how much the performance varied between folds.

2.5 Model Selection

The polynomial degree was selected based on the lowest mean validation MSE obtained through 5-fold cross-validation. For each degree, the mean validation MSE across the five folds was calculated. The degree having the minimum mean MSE was selected as the final polynomial degree. Evaluating the model across multiple polynomial degrees also helps identify potential underfitting and overfitting.

Using cross validation allows the degree to be selected based on its performance on unseen validation portions of the training data rather than its performance only on the data used for fitting. The final model was selected based on the lowest mean validation MSE obtained through 5-fold cross-validation. The selection was performed across all tested polynomial degrees, regularization types (L1 and L2), and α values. The combination with the minimum mean validation MSE was selected as the final model.

2.6 Final Model Training

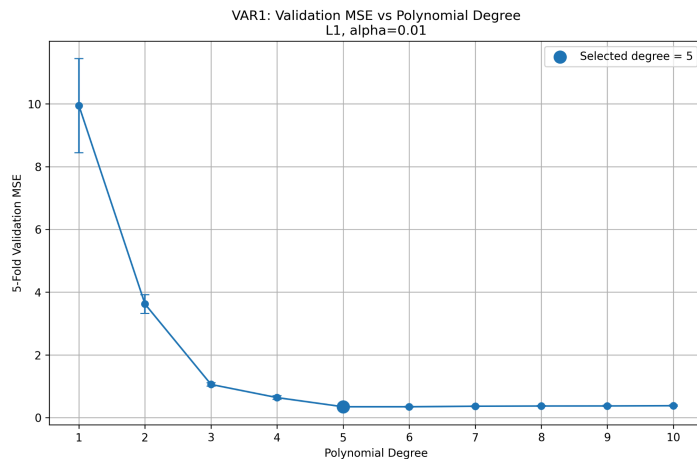
After selecting the polynomial degree using 5-fold cross-validation, the selected model was retrained using 100% of the available training data. This allows the final model to make use of all the available labelled training examples after the model-selection process has been completed.

The separate test dataset was then passed through the same polynomial transformation and the retrained regression model was used to generate the final predictions.

3. Question 1- Polynomial Regression

3.1 Experimental Results

The validation results for polynomial degrees 1–10 are shown below.



The mean validation MSE decreased from degree 1 to degree 5. Degree 4 achieved the lowest mean MSE of 0.34841935, after which the validation error increased.

3.2 Degree Selection

The selected polynomial degree for Question 1 is 4, as it gives the lowest mean validation

Degree = 5

Mean MSE = 0.34841935

Mean R2 = 0.96775462

3.3 Final Model and Prediction

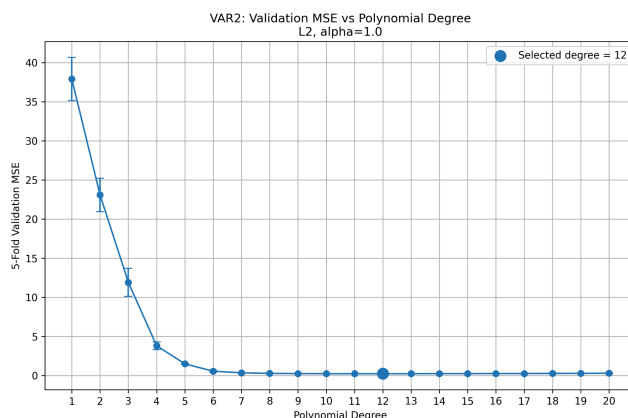
The degree-5 polynomial model was retrained using the complete Question 1 training dataset and then used to generate predictions for the test dataset.

Prediction file: IMT2024070_pred_var1.csv

4. Question 2 - Polynomial Regression

4.1 Experimental Results

The validation results for polynomial degrees 1–20 are shown below.



The validation MSE decreased substantially from degree 1 through degree 12. Degree 12 achieved the lowest mean validation MSE of 0.24068865. After degree 12, the MSE began to increase, with substantially larger errors at higher degrees.

4.2 Degree Selection

The selected polynomial degree for Question 2 is 12, as it gives the lowest mean validation MSE:

Degree = 12

Mean MSE = 0.24068865

Mean R2 = 0.99529253

4.3 Final Model and Prediction

The degree-12 polynomial model was retrained using the complete Question 2 training dataset and then used to generate predictions for the test dataset.

Prediction file: IMT2024070_pred_var2.csv

5.Overall Results

The final selected models are summarized below.

Question	Features	Degrees Tested	Selected Degree	Mean CV MSE	Mean CV R ²
Q1	x1-x6	1-10	5	0.34841935	0.96775462
Q2	x1-x3	1-20	12	0.24068865	0.99529253